

Requirement Analysis

Smart Lender: AI-Powered Loan Approval Prediction System

ID	Requirement
FR-1	Accept applicant details through a user-friendly web form (Gender, Marital Status, Dependents, Education, Self Employed, Applicant Income, Co-applicant Income, Loan Amount, Loan Amount Term, Credit History, Property Area).
FR-2	Preprocess the input data by handling missing values, encoding categorical variables, and applying feature scaling before prediction.
FR-3	Predict whether the loan application is Approved or Rejected using the trained Machine Learning model.
FR-4	Display the prediction result instantly through the Flask web application.
FR-5	Allow the trained model to be updated or retrained using new loan application data for improved prediction accuracy.

ID	Requirement
NFR-1	The system should generate loan predictions within 2 seconds .
NFR-2	The web interface should be simple, responsive, and easy to use for all users.
NFR-3	The prediction model should provide consistent and reliable results for similar input data.
NFR-4	The application should securely process applicant information without exposing sensitive data.
NFR-5	The system should be maintainable and allow future integration with cloud platforms or banking systems.

Data Requirements

The project uses the **Loan Prediction Dataset** obtained from Kaggle. The dataset contains historical loan application records with applicant details such as:

- Gender
- Marital Status
- Dependents
- Education
- Self Employed
- Applicant Income
- Co-applicant Income
- Loan Amount
- Loan Amount Term
- Credit History
- Property Area
- Loan Status (Target Variable)

The dataset is preprocessed by handling missing values, encoding categorical variables, balancing the data using **SMOTE**, and applying **StandardScaler** before training the Machine Learning models.

Hardware Requirements

- Processor: Intel Core i3 or above
- RAM: Minimum 4 GB (8 GB Recommended)
- Storage: Minimum 2 GB Free Space

- Internet Connection (Required for downloading packages and GitHub access)

Software Requirements

- Operating System: Windows 10/11
- Python 3.10 or above
- Jupyter Notebook
- Visual Studio Code
- Flask
- Scikit-learn
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Imbalanced-learn (SMOTE)
- Git & GitHub
- Web Browser (Google Chrome, Microsoft Edge, or Mozilla Firefox)

Technology Stack

Category	Technology
Programming Language	Python
Web Framework	Flask

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Frontend	HTML5, CSS3
Data Processing	Pandas, NumPy
Machine Learning	Scikit-learn
Dataset Balancing	SMOTE
Visualization	Matplotlib, Seaborn
IDE	Visual Studio Code, Jupyter Notebook
Version Control	Git & GitHub